

CLASS TEST-1
MML

Time: 30 minutes

Max. Points: 10

Here, we follow the same notations used in class.

- (1) **(1+1+1+1 Points)** Let $\mathbb{R}_+$ be the set of real numbers. Now define plus and scalar multiplication as follows

$$x \oplus y := x \times y, \quad c \otimes x := x^c.$$

Now $(\mathbb{R}_+, \oplus)$ is a vector space over $\mathbb{R}$ with above defined operations (You can assume this fact). Find the additive identity of this vector space. Is $\mathbb{R}_+$ finite dimensional over $\mathbb{R}$? If yes, find a basis and its dimension.

- (2) **(4 Points)** Prove or disprove e^x and e^{-x} are linearly independent over $\mathbb{R}$.
- (3) **(1+1+1+1 Points)** Let $T : (\mathbb{R}, +) \longrightarrow (\mathbb{R}_+, \oplus)$ be the map defined by $T(x) = e^x$. Prove or disprove T is a linear map. If yes, find its kernel, image and its rank.
- (4) **(1+ 1+ 2 Points)** Does there exist a linear map from $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ such that the image is generated by $(1, 2, 3), (1, 4, 9)$? If yes, give an explicit map? Further how many such linear map will be?